Deep Learning for NLP

Large Language Models (LLMs)
DeepSeek Reasoning

Query

The denominator of a fraction is 7 less than 3 times the numerator. If the fraction is equivalent to 2/5, what is the numerator?

Correct reasoning trace

Step 1	*Let the numerator be x.*
Step 2	*The denominator is 3x-7.*
Step 3	*We know that x/(3x-7) = 2/5.*
Step 4	*Therefore, 5x = 6x-14.*
Step 5	*Finally, we get x= 14.* (Correct)

Learning goals

- understand DeepSeek reasoning as an example of state-of-the-art reasoning

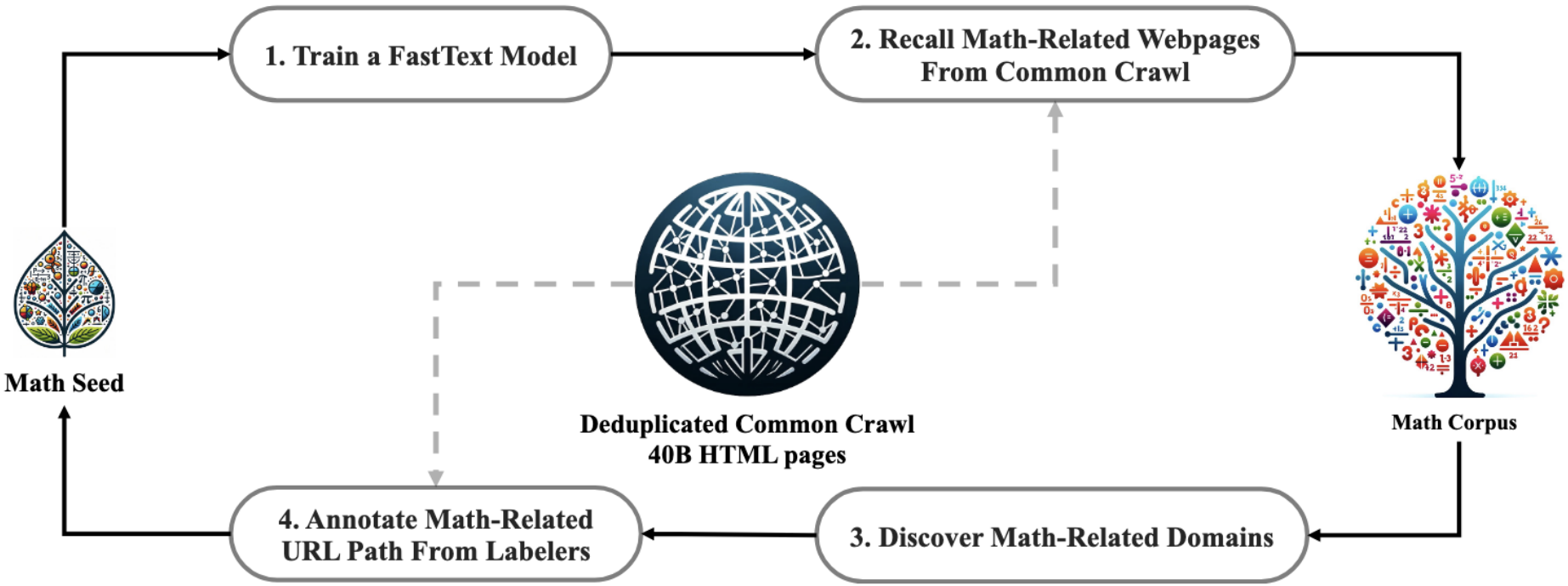
LM REASONING IN 2026

- Instead of using prompting, LMs are trained to reason.
- Supervised finetuning (SFT)
- Reinforcement learning (RL)
- Training data:
  - Complex pipeline to create reasoning-relevant web corpus
  - Data generated by LM, then automatically labeled
- Motivation 1: Instead of requiring prompts, LMs should become native reasoners.
- Motivation 2: Human-curated data limit the reasoning power. RL may discover novel and more diverse ways of reasoning.
- Motivation 3: Inference-time scaling. chain-of-thought few-shot prompting fixes the length, not applicable to really hard problems that require flexible reasoning.

SOURCES

- First major open reasoning LM (as opposed to proprietary closed): DeepSeek
- ▸ DeepSeek-AI, 2025
- ▸ Nature, 2025

DATA FOR SFT: 40B TOKENS! I



▸ DeepSeekMath, 2024

REINFORCEMENT LEARNING I

- In reinforcement learning, an agent takes a sequence of actions towards a goal and gets feedback on the success of that sequence at the end.
- Details: Next lecture
- For reasoning trajectories, the feedback is: correct or incorrect.
- (thanks to Emma Brunskill)

	supervised learning	reinforcement learning
goal	optimal predictor	optimal policy
learn from experience?	yes	yes
generalization?	yes	yes
feedback	rich (each token)	sparse (in/correct)
feedback when?	immediately	delayed
exploration?	no	yes

REASONING TRACES: FORMAT

- The User asks a question and the Assistant solves it.
- The Assistant first thinks about the reasoning process “in its mind” and then provides the answer to the User.
- The reasoning process and answer are enclosed within <think>...</think> and <answer>...</answer> tags
- `<think> reasoning process here </think>`
- `<answer> answer here </answer>`
- “We intentionally limit our constraints to this structural format, avoiding any content-specific biases to ensure that we can accurately observe the natural progression of the model during the RL process.”
- So this is not a few-shot approach!

EXAMPLE OF A REASONING TRACE I

Query

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Question:

Is this chain of thought few-shot prompting?

AUTOMATIC, NON-HUMAN REWARDS FOR LM-GENERATED REASONING TRACES

Accuracy rewards evaluate whether the response is correct. For example, in the case of maths problems with deterministic results, the model is required to provide the final answer in a specified format (for example, within a box), enabling reliable rule-based verification of correctness. Similarly, for code competition prompts, a compiler can be used to evaluate the responses of the model against a suite of predefined test cases, thereby generating objective feedback on correctness.

AUTOMATIC VERIFIER/VERIFIABLE REWARDS: AIME EXAMPLE I

PROMPT

Let $b \geq 2$ be an integer. Call a positive integer n b-*eautiful* if it has exactly two digits when expressed in base b, and these two digits sum to $\sqrt{n}$. For example, 81 is 13-eautiful because $81 = \underline{6}\ \underline{3}_{13}$ and $6 + 3 = \sqrt{81}$. Find the least integer $b \geq 2$ for which there are more than ten b-eautiful integers.

Please reason step by step, and put your final answer within \boxed{}.

Evaluation

Parse the final answer within \boxed{} and use a rule-based grader to determine if it equals the ground truth. Round numerical values as needed, and use 'SymPy' to parse expressions.

INFERENCE TIME SCALING

- For a hard problem, it often helps to think for a long time to come up with the solution.
- Inference time scaling: teach the model to think for a while before giving the answer.
- Here, thinking means that the model produces potentially long output, the reasoning trace, that leads to the correct answer.

WHAT ELEMENTS OF REASONING ARE LEARNED?

- dynamic strategy adaptation
- self-verification
- reflection
- backtracking
- reasoning traces are getting longer
- exploration of alternative approches
- diverse and sophisticated reasoning behaviors

AHA MOMENT I

Question: If $a > 1$, then the sum of the real solutions of $\sqrt{a - \sqrt{a + x}} = x$ is equal to

---

Response: <think>

To solve the equation $\sqrt{a - \sqrt{a + x}} = x$, let's start by squaring both $\cdots$

$\left(\sqrt{a - \sqrt{a + x}}\right)^2 = x^2 \implies a - \sqrt{a + x} = x^2.$

Rearrange to isolate the inner square root term:

$(a - x^2)^2 = a + x \implies a^2 - 2ax^2 + (x^2)^2 = a + x \implies x^4 - 2ax^2 - x + (a^2 - a) = 0$

$\cdots$

Wait, wait. Wait. That's an aha moment I can flag here.

Let's reevaluate this step-by-step to identify if the correct sum can be $\cdots$

We started with the equation:

$\sqrt{a - \sqrt{a + x}} = x$

First, let's square both sides:

$a - \sqrt{a + x} = x^2 \implies \sqrt{a + x} = a - x^2$

Next, I could square both sides again, treating the equation: $\cdots$

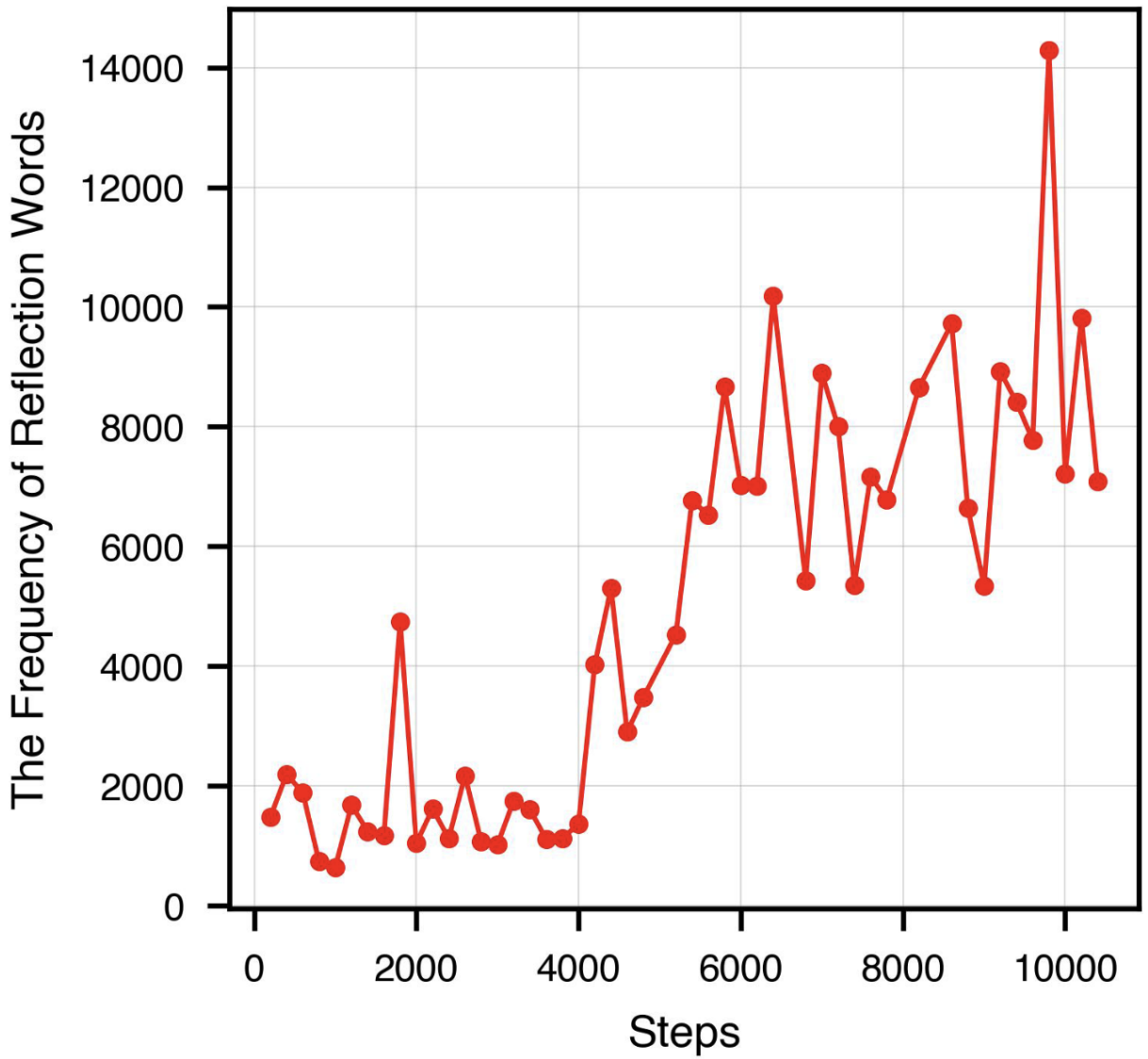
ANOTHER EXAMPLE FOR AHA MOMENT I

How many r in raspberry? **Question**

Let's break down the process of counting the letter 'r' in the word "raspberry" ...
* First letter: 'r' - This is an 'r', count = 1.
* Second letter: 'a' - Not an 'r', count remains 1 ...
* Sixth letter: 'e' - Not an 'r', count remains 1.
* Seventh letter: 'r' - This is an 'r', count = 2.
* Eighth letter: 'y' - Not an 'r', count remains 2 ...
The number of 'r's in "raspberry" is 2.
Wait, let's re-read the question carefully. It asks "How many r in raspberry?" ... * r - a - s - p - b - e - r - r - y ... * First 'r' ... * Second 'r' ... * Third 'r' ... Count = 3 ... **Reasoning trace**

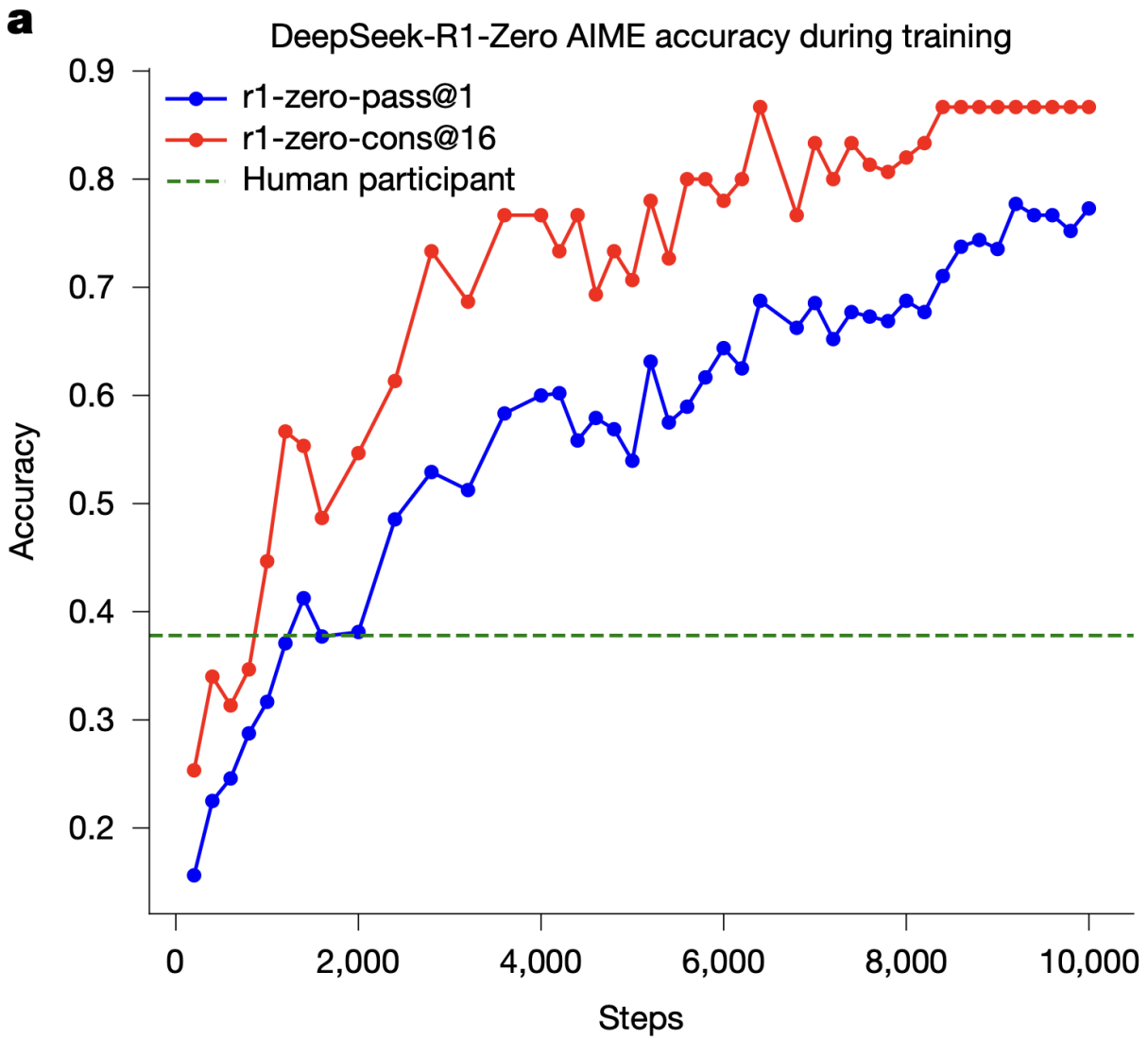
My initial answer of 2 was incorrect due to a quick reading of the word. **Final Answer:** The final answer is $\boxed{3}$ **Response**

THE RISE OF REFLECTIVE TERMS I

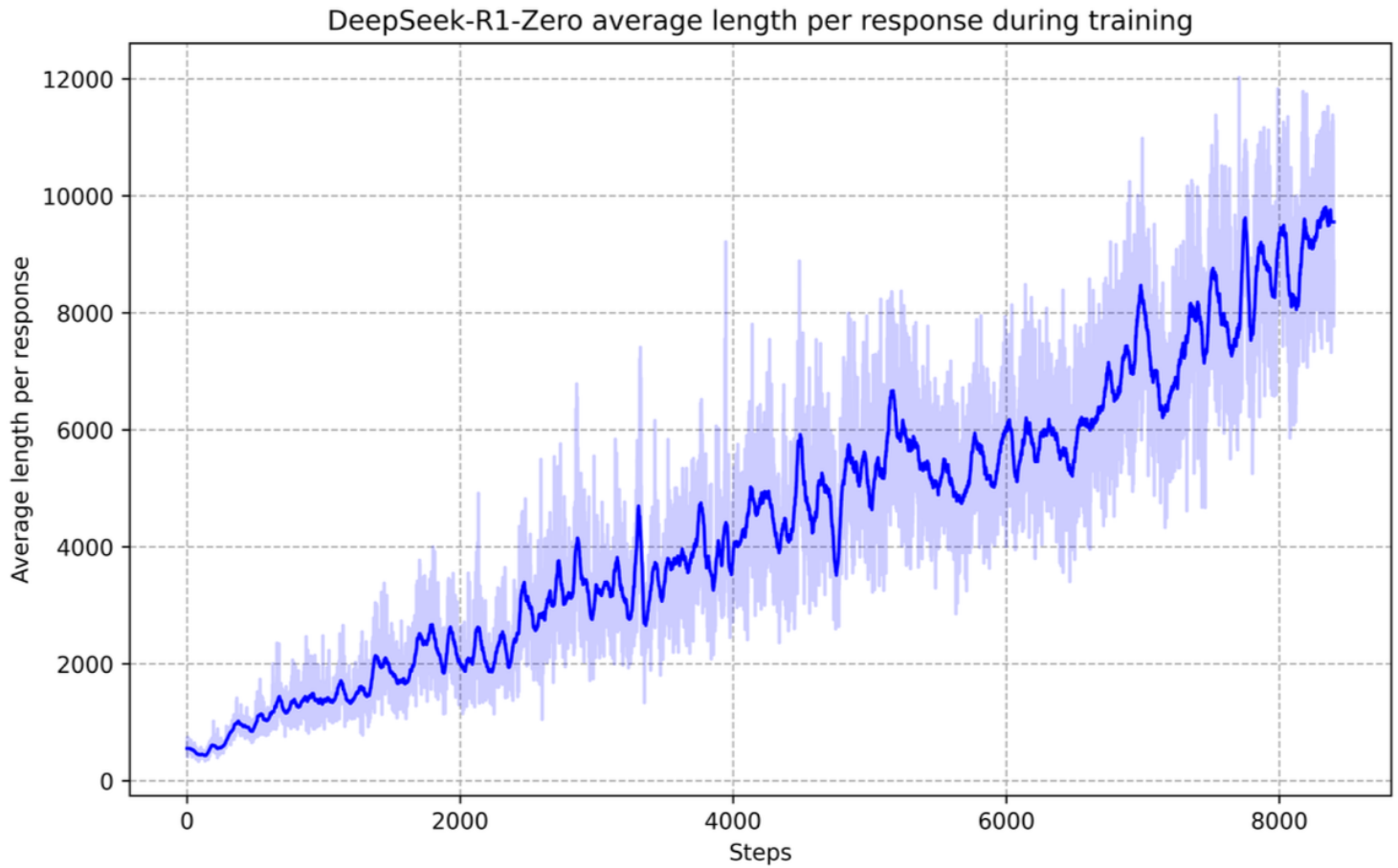


Frequency of reflective terms (‘wait’, ‘mistake’, ‘however’, ‘but’, ‘retry’, ‘error’, ‘verify’, ‘wrong’, ‘evaluate’, ‘check’) in model output during training

ACCURACY AIME: ABOVE HUMAN AVERAGE I



R1 SCALING: WITHOUT EXPLICIT LENGTH SIGNAL I



UNFAITHFULNESS OF LMS TO TRUE REASONING I

```
Human: Answer in one word. What is 36+59?

Assistant: 95

Human: Briefly, how did you get that?

Assistant: I added the ones (6+9=15), carried the 1, then added the tens
(3+5+1=9), resulting in 95.
```

UNFAITHFULNESS OF LMS TO TRUE REASONING
I

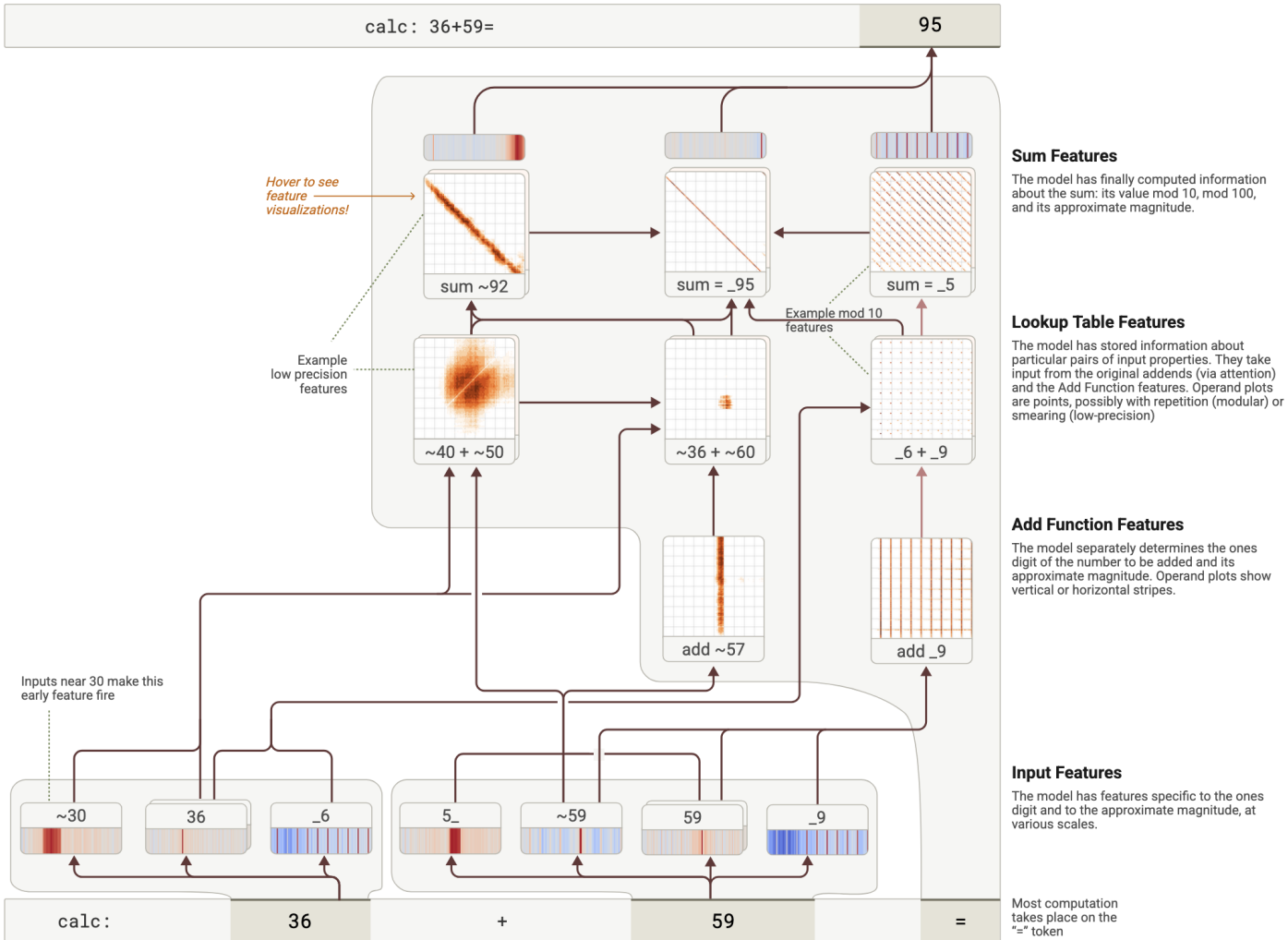


Figure 26: A simplified attribution graph of Haiku adding two-digit numbers. Features of the inputs feed into separable processing pathways.

THINGS WE HAVE NOT COVERED

- use an eco system of several models:
  for distillation, generation, verification
- some “light” human post-editing of training data
- rejection sampling:
  keep only high-quality, high-confidence training data
- apart from math, they also do this for code and other domains
- major hardware optimization work
- claim: cost of training $<$ 1 000 000 million dollar

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